

A DFT/machine-learning hybrid method for the prediction of $^3J_{\text{HCCH}}$ couplings

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Abstract

A machine learning model for the prediction of vicinal proton–proton couplings has been developed based on a hybrid representation that includes geometrical and electronic parameters derived from natural bond orbital (NBO) analysis of low-level BLYP/STO-3G computations. The model can predict $^3J_{\text{HH}}$ couplings with accuracy comparable or better than the well-known Altona equation, and it can provide sensible $^3J_{\text{HH}}$ predictions in systems not well handled by the Altona equation such as epoxide or cyclopropane rings.

KEYWORDS

DFT, machine learning, prediction, vicinal couplings

1 | INTRODUCTION

The prediction of nuclear magnetic resonance (NMR) parameters, through discrete Fourier transform (DFT) computations, has become a widespread tool in NMR-based structural elucidation of organic compounds.^[1–3] Use of quantum chemical predictions of chemical shifts and scalar couplings allows to discriminate not only constitutional but also configurational isomers with, in many cases, very high degrees of confidence.^[4] Particularly, DFT predictions of scalar couplings allow to go well beyond the limitations imposed by the Karplus-type equations^[5] either in terms of types of coupled pairs that can be computed or in terms of accuracy.

However, the prediction of J couplings is usually computationally more expensive than that of the chemical shifts. Nevertheless, for practical structural elucidation purposes, the computational time can be greatly reduced because for couplings between light atoms, computation of the Fermi contact term is usually necessary because the spin-dipole and spin-orbital terms are much smaller in magnitude. Even more, the paramagnetic and

diamagnetic parts of the spin-orbital term tend to cancel each other. Also, specialized basis sets and the use of empirical calibrations^[6,7] may result in important savings of computational time, as can be seen in the various contributions from Kutateladze group,^[8] making J -coupling prediction a very practical tool for structure elucidation.^[9]

In the last few years, we have witnessed an important increase in the use of the so-called machine learning (ML) methodologies for fast prediction of molecular properties from three-dimensional structure.^[10] Although the NMR field has a long tradition of different ML techniques^[11] for the prediction of chemical shifts and/or scalar couplings, they are mostly devoted to prediction from a 2D graph, therefore averaging or making implicit conformational effects on the predicted NMR parameters.^[12,13] Clearly, extension of computer-assisted structure elucidation (CASE) systems to determination of configuration/conformation needs to take into account three-dimensional information in the core of the NMR parameter prediction. Therefore, the development of ML NMR predictors based on three-dimensional

structure^[14,15] can have a very important impact in the performance and automation of 3D structure of NMR determination protocols because ML algorithms can easily be orders of magnitudes faster than current DFT methodologies. Note that hybrid DFT/ML methods where the ML part tries to improve the accuracy^[16] or statistical significance^[17] of the DFT NMR prediction are also already known.

Very interestingly, recent important contributions have been reported for the fast prediction of chemical shifts from a molecular three-dimensional structure.^[18–20] Additionally, prediction of $^1J_{\text{CH}}$ couplings was also reported by the group of Butts.^[20] A common particularity in all of these contributions is that the ML protocols were trained not against experimental data but against a selected DFT level of theory. This is because training against experimental data would require proper conformational deconvolution of the experimental data that would make difficult to establish firm correspondences between particular conformations and shifts. In this respect, the target of these studies is not the reproduction of experimental shifts but to match a particular level of theory that can be considered of enough accuracy to be later used for practical applications.

The main problem, perhaps, in the deployment of ML protocols for molecular properties prediction from three-dimensional structure is to choose a proper unique molecular representation that can be understood by the chosen computational algorithm. One of the simplest representations is the so-called Coulomb matrix^[21] that encodes the electrostatic nuclear potential in a rotation invariant way, but other more sophisticated representations are available as can be seen in the work of Emsley and Butts.

On the other hand, nowadays, through improvements in hardware and algorithms such as the use of resolution of the identity,^[22] DFT computations have become cheaper and cheaper. It is our working hypothesis that a combination of local geometric parameters and electronic parameters such as atomic charges and hybridization states obtained by analysis of very low-level and fast DFT calculations could be used as a molecular representation for an ML system capable to reproduce the results of high-level DFT predictions but with a very important reduction in computing time. As a particular application, we will try to implement an ML system capable to reproduce the results of DFT “high-level” predictions of vicinal $^3J_{H(Csp^3)(Csp^3)H}$ couplings, the most used ones for practical applications. The ML predicted coupling as such will have enough accuracy as to be used for practical structural elucidation purposes hopefully surpassing the performance and applicability of the vastly used Altona equation.^[23]

2 | METHODOLOGY

The first 6,000 entries in the PubChem NIH database^[24] (<https://pubchem.ncbi.nlm.nih.gov>) were retrieved as MMFF94^[25] optimized structures^[26] using an in-house written Python3 script available in the supporting information. The same script, which made use of the Open Babel^[27] 3.0 Python3 bindings, was used to filter the structures according to the following criteria: (i) molecule should not be charged; (ii) molecular weight should not be higher than 500 Da; and (iii) molecules should contain only atoms from hydrogen to fluorine. The first 5,000 entries resulted in a total of 19,955^[28] couplings. These couplings were used as the training set. Entries from 5,001 to 6,000 furnished a total of 4,499 coupling that were used as the test set (note that the PubChem CID entry 5836 and its methoxy derivative CID 5980 have to be excluded from the test as incorrect geometries were generated by the PubChem 3D engine). A small test set, with 43 $^3J_{\text{HH}}$ couplings, was comprised from the strychnine, α -erythroidine, and ajmaline natural alkaloids (PubChem CID entries 441071, 441076, and 441080). For cross-validation purposes, an alternative training system was built using the last 19,955 couplings and tested on the first 4,499 ones.

Two single-point DFT computations were made on the MMFF94 retrieved geometries. The first one, from here called *low* level, was a single-point BLYP^[29,39] DFT computation using the minimal STO-3G basis set. In this low-level computation, atomic charges and hybridization states were calculated by NBO analysis.^[28] Because natural bond orbital (NBO) gives the percentage of *s*, *p*, and *d* character of particular bonds, the hybridization states were averaged all over the bonds of a particular atom and represented as the percentage of *s* character. The second *high* level of computation used the B3LYP functional^[31] and the 6-31G* basis set. At this level, coupling constants were computed using only the Fermi contact term (`nmr=onlyfc` Gaussian09 keyword). All DFT computations were done using the Gaussian09 package that also includes the NBO 3.1 program.^[32] For some illustrative systems, $^3J_{\text{HH}}$ couplings were also computed using the Altona^[23] equation as implemented in the MSpin-JCoupling package.^[33] The C++ programming interface of this package, along with the LMfit implementation^[34] of the Levenberg–Marquardt algorithm,^[35,36] was also used to implement a small C++ program to reparametrize the Altona equation in order to fit the computed couplings of the training set.

The ML kernel ridge regression^[37] computations were performed using in-house made Python3 scripts that employed Lilienfeld's group QML Python3

ML package^[38] linked against OpenBLAS libraries (<http://www.openblas.net>).^[39]

3 | RESULTS AND DISCUSSION

Our working methodology was based on the hypothesis that local electronic parameters such as atomic charges and hybridization states will encode an important number of local structural features but also non-local effects, such as mesomerism, which can affect the magnitude of the scalar coupling. Complemented by local geometrical parameters, such as of course the value of the proton–proton dihedral angle, bond angles, and bond lengths, we hoped that these parameters could be used to build a simple molecular representation that could be used in an ML protocol to mimic the results obtained within a chosen DFT method.

Because one of our principal goals in our research is to achieve a higher as possible degree of automation and computational performance in the structure elucidation process,^[14] it is clear that the electronic parameters should be retrieved from a lowest possible level of theory as possible. In our case, we chose a non-hybrid GGA functional and a minimal basis set (STO-3G). Although not used during this work, it should be noted that BLYP computations are susceptible of very important speed-ups through the use of the resolution of the identity approximation. Also, DFT computations are directly done on the MMFF94 molecular mechanics geometries as DFT optimizations will have an important associated computational cost.

It is well known in the computational chemistry field that W NBO methodology^[28] can provide chemically meaningful atomic and bond parameters, such as atomic charges, hybridization states, and bond orders, from computed self-consistent field (SCF) densities. NBO analysis was conducted for all the molecules at the BLYP low level of theory using the NBO 3.1 program as deployed in Gaussian09. The BLYP computations took approximately 1–2 min of computation time per structure using four cores.

Note that, similarly to previous work on prediction of chemical shifts, already discussed in Section 1, instead of training our machine to fit available experimental data, we decided to mimic results obtained through DFT computations. Obviously, this introduces a limit on the accuracy of the predicted couplings as, for instance, vibrational effects are neglected. However, this allows to overcome the problem of experimental determination of the couplings. Extraction of accurate proton–proton couplings can be, in many circumstances, hampered by overlapping and strong-coupling effects. Additionally, in most

of the cases, they represent conformationally averaged values and therefore cannot be directly correlated to a simple three-dimensional structure.

We chose as our high level of theory the B3LYP/6-31G* as it allows to compute sufficiently accurate couplings with minimum computational expenses. In any case, the ML protocol can be trained with data obtained at any level of theory. Even more, the computed values, or their ML counterpart, can be a posteriori corrected by the use of calibrated scaling factor (see above) if desired.

Our molecular representation was chosen as a combination of geometric and electronic parameters involving the atoms in the coupling path H1–C1–C2–H2. The first geometric ones were chosen in order to take into account the geometric nature of the coupling path.

- p₁. The cosine and sine of the H–H1–C1–C2–H2 dihedral angle;
- p₂. The three bond distances H1–C1, C1–C2, and C2–H2;
- p₃. The proton–proton distance H1–H2; and
- p₄. The cosines of the two H–C1–C2 and H2–C2–C1 angles.

Additional geometric parameters took into account the orientation, geometry, and nature of α -substituents X,^[23]

- p₅. The cosine and sine of the dihedral angles X1–C1–C2–H2, X2–C1–C2–H2, and X3–C2–C1–H1 and X4–C2–C1–H1;
- p₆. The cosines of the X1–C1–X2 and X3–C2–X4 angles; and
- p₇. The distance between the carbon atoms and the α -substituents X1–C1, X2–C1, C2–X3, and C2–X4.
- p₈. Hybridization states (represented as *s* character) of the H1–C1–C2–H2 atoms;
- p₉. Atomic charges of the H1–C1–C2–H2 atoms;
- p₁₀. Atomic charges of the α attached atoms; and
- p₁₁. Hybridization states (*s* character) of the α attached atoms.

named as X1–C1–X2 and X3–C2–X4. These parameters are The following electronic parameters, as obtained from the NBO B3LYP/STO-3G analysis, were incorporated into the representation. The above representation was called model ML-1. We also built a representation where NBO parameters were dropped but instead information on the α -substituents was incorporated as the natural logarithm of the atomic number (p₁₂ parameter). This model was called ML-2.

We also played with the incorporation of a local Coulomb matrix^[40] to the representation. This Coulomb matrix was built placing the origin of coordinates not in a particular atom but in the middle of the C–C bond. The diagonal and off-diagonal elements of the matrix were then computed according to the following expressions:

$$M_{ij} = Z_i^{2.4}; M_{ij} = \frac{Z_i Z_j}{\left(\|\mathbf{R}_i - \mathbf{R}_0\|_2 + \|\mathbf{R}_i - \mathbf{R}_j\|_2 + \|\mathbf{R}_i - \mathbf{R}_j\|_2 \right)^\alpha} \quad (1.1)$$

where Z_i and Z_j ^[37] are the atomic numbers of the i th and j th atoms and $\mathbf{R}_i$ and $\mathbf{R}_j$ are their positions with respect to the origin of coordinates $\mathbf{R}_0$ placed at the middle of the C–C bond. Entries were ordered according to the distance of the i th atom with respect to $\mathbf{R}_0$. Note that we limited the size of the Coulomb matrix representation to a cut-off distance of 5 Å and to thirty atoms. If less than thirty atoms were present inside the cut-off radius, the Coulomb matrix representation was padded with zeros. The α parameter, responsible for screening of long-range interaction, was chosen as $\alpha = 3.0$. The Coulomb matrix representation was further scaled by a factor of 50 to keep its norm close to unity.

Our ML protocol was based on the kernel ridge regression,^[37] a method already used with success in the above mentioned work on chemical shift prediction. We employed a using a Gaussian kernel with Fröbenius norm as implemented in the QML package. The desired property is obtained from a particular $\tilde{\mathbf{X}}$ representation of the coupling through the expression:

$$^3J(\tilde{\mathbf{X}}) = \sum_i \alpha_i K(\tilde{\mathbf{X}}, \mathbf{X}_i), \quad (1.2)$$

where the kernel elements K_{ij} are computed as follows:

$$K_{ij} = K(\mathbf{X}_i, \mathbf{X}_j) = \exp\left(-\frac{\|\mathbf{X}_i - \mathbf{X}_j\|_2^2}{2\sigma^2}\right). \quad (1.3)$$

The regression coefficients α_i are then obtained through

$$\alpha = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{J}, \quad (1.4)$$

being $\mathbf{I}$ and $\mathbf{J}$ the unit matrix and the vector of DFT computed values on the training set, respectively.

Length-scale σ and normalizing parameter λ were optimized using a simple grid searching, with σ and normalization parameter λ chosen taken from the (0.25, 0.5,

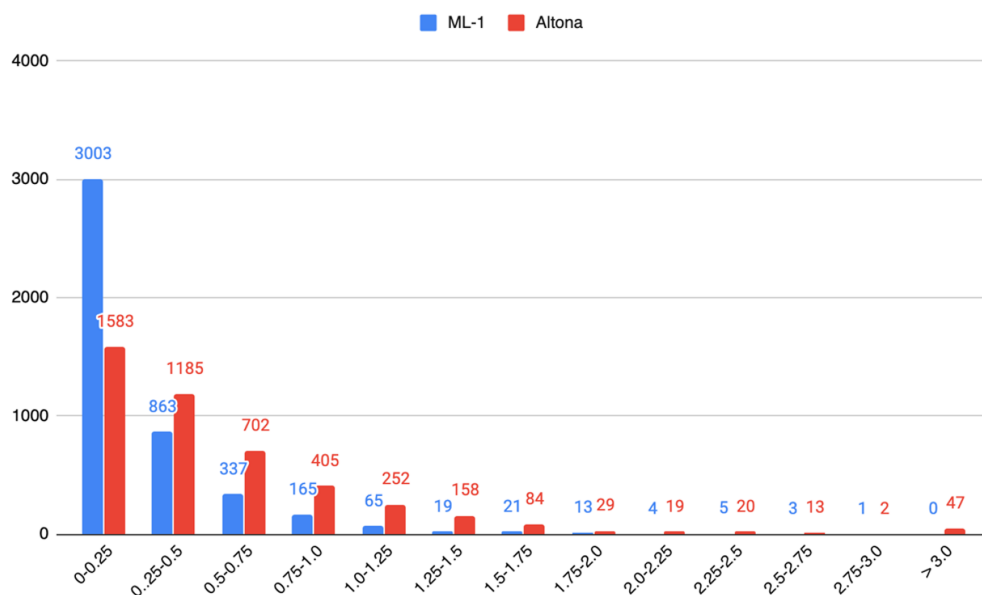
0.75, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5) and (10^{-5} , 10^{-4} , 10^{-3} , 10^{-2} , 10^{-1} , 1) sets, respectively. The optimal σ and λ values were taken in order to minimize the root mean square error (RMSE) between the predicted and DFT computed values.

The best results for the NBO-based ML-1 model were obtained for σ and λ values of 1.0 and 0.001, respectively. These values corresponded to an RMSE of 0.39 Hz and a largest absolute error of 2.84 Hz. From the 4,499 couplings in the large test set, 4,368 couplings (97.1%) were predicted with a deviation of less than 1 Hz, whereas 118 (2.6%) had deviations of 1–2 Hz, and only 13 (~0.3%) corresponded to deviations larger than 2 Hz (Figure 2). A simple cross-validation was done by taking the 4,499-coupling test set from the first structures rather than the last ones and using the remaining couplings as the tests set. Somewhat unexpectedly, a much larger RMSE was obtained (0.47 Hz). The reason for this large RMSE was the presence on the test set of a cyclohexadienyl structure with unusually large $^3J_{\text{HH}}$ couplings. This issue will be further discussed below. Nevertheless, the distribution of deviations was very similar with 97.1% of predicted couplings having deviations below 1 Hz and 2.5 below 2 Hz (see Figure S1).

How important is the use of the DFT/NBO electronic parameters for the quality of the prediction? To answer this question, we tested the ML-2 model where charges and hybridization states information (parameters p_8 to p_{11}) were left out. In order to incorporate some information on the nature of the α -atoms, we included the logarithm of the atomic number of these atoms as an extra parameter (p_{12}). The best result (σ and λ values of 1.0 and 0.01) furnished a significantly higher RMSE (0.420) than the ML-1 model, and more importantly, the number of outliers ($|J_{\text{DFT}} - J_{\text{ML}}| > 2.0$ Hz) increased significantly with 22 values over those 13 values on the ML-1 model (Figure 1).

Evidently, our machine will be useful if it can outperform Altona equation predictions. It must be taken into account that Altona equation was parametrized to reproduce experimental values rather than DFT computed ones. In order to do a fair comparison of our ML models against Altona equation, we reparametrized Altona's original set of parameters, named as B, D, and E in the original article, against our large test set of 4,499 DFT couplings. These B, D, and E parameter sets were meant to be applied to disubstituted, three substituted, and tetrasubstituted cases, respectively. Computed parameters can be seen in the supporting information. Computing $^3J_{\text{HH}}$ couplings using this DFT-based parametrized Altona equation resulted in an RMSE of 0.81 Hz, significantly higher than that of the ML-1 model. More importantly, the number of outliers was significantly

FIGURE 2 Number of couplings in the large test set for increasing absolute error for the machine learning (ML)-1 model (hyperparameters: left: $\sigma = 1.0$, $\lambda = 0.001$) and the Altona (discrete Fourier transform parametrized) equation

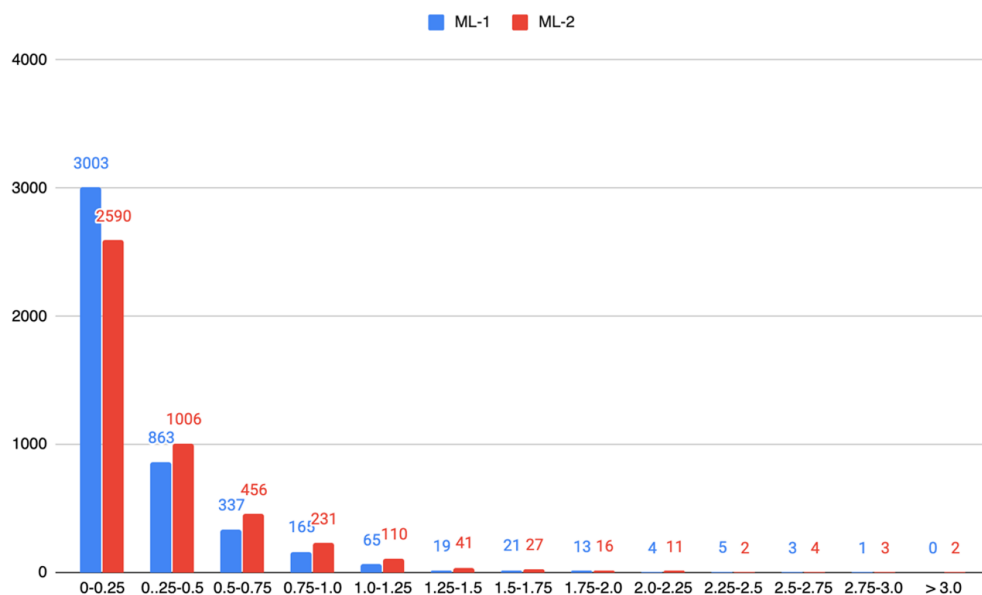


larger than in our ML-1 method as can be seen in Figure 2. Taking a look at the nature of the outliers in the Altona prediction, these were mostly systems such as cyclopropyl rings or those with important hyperconjugative effects of C–H bonds with unsaturated bonds. A glimpse of the more important outliers in the Altona predictions can be seen in Figure 3 along with the ML predicted values. Not only the ML-1 model but also the ML-2 model, which does not require any DFT computation, provided sensible predictions for those difficult systems.

When the above optimized ML-1 model was applied for prediction of the $^3J_{HH}$ couplings in our small test set of natural products, we obtained an RMSE of 0.65 Hz

and a largest deviation of 2.29 Hz. Five couplings over the 43 coupling sets have absolute deviations over 1 Hz and only one over 2 Hz. Predicted values for strychnine can be seen in Table 1. For this set, the DFT-parametrized Altona equation furnished an RMSE of 0.68 Hz with one coupling deviation over 1.0 Hz and another one over 2.0 Hz. Corresponding predicted values for α -erythroidine and ajmaline can be checked in the supporting information. For the sake of comparison, we present also the predicted Altona values in the tables, using the standard as well as the DFT-based parametrization. Very interestingly, the ML-2 model performed better for this subset with an RMSE of 0.48 Hz and a largest deviation of 1.2 Hz, with only three couplings presenting

FIGURE 1 Number of couplings in the large test set for increasing absolute error for the machine learning (ML) models ML-1 and ML-2



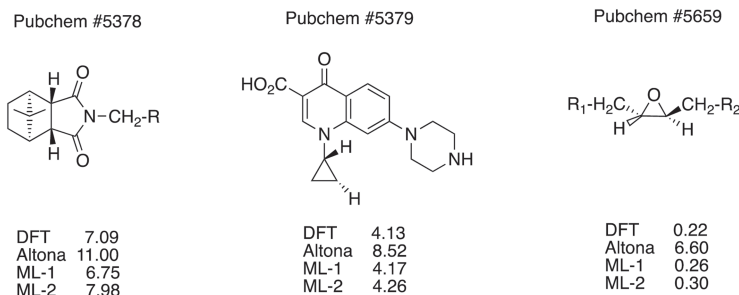


FIGURE 3 Principal outliers in the test set for Altona equation-based predictions (discrete Fourier transform [DFT]-based parametrization) and machine learning (ML)-1 and ML-2 predicted values. All values in Hz

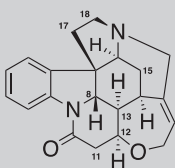
deviations over 1.0 Hz barely above. Clearly, the ML predicted values are sufficiently close to the DFT ones as to be applied to structural elucidation purposes of common natural products.

It should be noted that including local Coulomb matrixes in the representation, as explained in the methodology section, did not result in better predictive models. The best σ and λ parameters were 2.0 and 0.01,

respectively. The obtained RMSE and large deviation were 0.48 and 3.84 Hz, respectively. The distribution of absolute value deviations also significantly worsened as can be seen in the SI (Figure S2). Since there were no hints that Coulomb matrixes could ameliorate our model we did not further continue working on this direction.

Finally, a glimpse on the range of DFT computed values in the training set gave us a little surprise. We

TABLE 1 Predicted $^3J_{\text{HH}}$ couplings for strychnine

						
3J coupling	Exp. ^[41]	B3LYP/6-31Gc	Altona ^[a]	Altona ^[b]	ML-1	ML-2
H8–H13	10.48	9.63	9.86	9.02	10.81	9.44
H11 α –H12	8.43	7.94	9.71	10.78	8.80	9.04
H11 β –H12	3.25	2.28	3.72	3.22	3.74	3.90
H12–H13	3.45	2.93	3.35	3.60	3.59	3.42
H13–H14	3.05	2.87	3.05	2.82	3.96	3.81
H14–H15 α	1.86	1.75	2.19	2.13	1.90	1.79
H14–H15 β	4.78	4.61	4.37	4.69	4.59	4.20
H15 α –H16	2.16	1.96	2.60	2.66	2.59	2.07
H15 β –H16	3.86	3.68	3.90	3.98	3.44	3.59
H17 α –H18 α	**** ^d	7.85	7.87	7.98	7.85	7.85
H17 α –H18 β	**** ^d	11.12	10.68	10.81	11.03	10.87
H17 β –H18 α	**** ^d	0.57	0.42	0.75	0.41	0.36
H17 β –H18 β	**** ^d	6.40	7.30	6.71	6.32	6.10
RMSE ^[c] (Hz)	—	—	—	0.47	0.70	0.50

Abbreviations: ML, machine learning; RMSE, root mean square error.

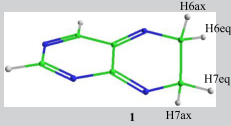
^aStandard parametrization.

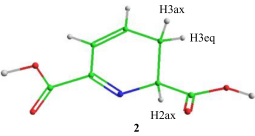
^bDFT-based parametrization.

^cRMSE between strychnine DFT B3LYP/6-31* values and Altona/ML predicted ones.

^dNot experimentally available.

TABLE 2 Predicted scalar couplings for the 6,7-dihydropteridine system

					
		ML ^a $\sigma = 1.0$ $\lambda = 0.001$	ML ^a $\sigma = 1.0$ $\lambda = 1e-5$	ML-1 ^b $\sigma = 1.0$ $\lambda = 0.001$	ML-1 ^b $\sigma = 1.00$ $\lambda = 1e-5$
B3LYP/6-31G* ^a					
$^3J_{6ax,7ax}$	29.04	12.73	14.00	—	—
$^3J_{6ax,7eq}$	7.18	3.69	3.97	—	—
$^3J_{6eq,7ax}$	8.49	3.71	4.00	—	—
$^3J_{6eq,7eq}$	2.39	1.81	1.78	—	—

					
		ML ^a $\sigma = 1.0$ $\lambda = 0.001$	ML ^a $\sigma = 1.0$ $\lambda = 1e-5$	ML-1 ^b $\sigma = 1.0$ $\lambda = 0.001$	ML-1 ^b $\sigma = 1.00$ $\lambda = 1e-5$
$^3J_{2ax,3ax}$	19.39	12.03	12.22	12.22	14.62
$^3J_{2ax,3eq}$	8.52	5.03	4.98	4.68	5.33

Abbreviation: ML, machine learning.

^aTraining set did not include entry 167 (**1**).

^bEnlarged training set including entry 167 (**1**).

observed a computed giant vicinal coupling of 29 Hz for 6,7-dihydropteridine **1** (PubChem entry 167), a reactive intermediate in the oxidation of tetrahydropteridine to 7,8-dihydropteridine (Table 2).^[42] During the revision of this manuscript, we showed that these large pathological couplings and similar systems are rooted in large hyperconjugative effects of the C–H bonds with the quinonoid system. Our computational study also showed that even simple cyclohexadienyl systems should also show rather large values with an axial–axial couplings close to 20 Hz.^[43]

How efficient would be our ML system to reproduce these abnormally large couplings? When running our cross-validation calculations of the ML-1 model, we noticed that the larger root mean square error (RMSE) in the cross-validation calculations was produced by a few outliers due to misprediction of the $^3J_{HH}$ couplings in the before mentioned 6,7-dihydropteridine system **1** and also the dihydropyridine system **2** (PubChem entry 17) as can be seen in Table 2. Reducing the smoothness of the kernel by setting the normalization parameter λ to 10^{-5} improved the results very marginally. Can we get a better prediction in **2** if the related

system **1** is included in the training system? We enlarged the training system to do so. Only when using a small kernel width $\lambda = 10^{-5}$ prediction was slightly improved. It seems therefore that our representation is unable to capture the structural details necessary to reproduce such abnormal couplings.

4 | CONCLUSIONS AND PERSPECTIVES

The present work shows that an ML predictor, developed through the kernel ridge regression procedures and based on the use of a molecular representation geometric and low-cost DFT-derived electronic parameters, can mimic “high-level” DFT computations of $^3J_{HH}$ couplings with sufficient accuracy as to be useful for structural elucidation of organic molecules while keeping an affordable computational effort. The model perform so well, if not better than the Haasnoot–Altona equation in common situations and, very importantly, can very well handle demanding systems such as cyclopropyl or epoxide rings. The present

model may have still large room for improvement through amelioration of the model, as well as a more extensive training. Interestingly, a simpler model (ML-2) only based on geometrical parameters and α -substituent atomic numbers, therefore not needing DFT calculations, presented also a good performance. Although the model presented here was trained in relatively low-level DFT computations, it can be trivially trained using higher computational levels that may involve the use of specialized basis set and the computations of spin-orbital J -coupling terms. Work is in progress to develop similar ML models of other structurally useful types of scalar couplings such as $^1J_{\text{CH}}$ couplings^[20,44] or long-range $^2J_{\text{CH}}$ and $^3J_{\text{CH}}$ couplings.^[45–47]

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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